Mensuration – 2D

Any triangle

$$\textbf{Area} = \frac{1}{2} \times \textbf{Base} \times \textbf{Height}$$

Equilateral triangle

Side = a

Perimetre = 3a

Area $= \frac{\sqrt{3}}{4} a^2$

Height $= \frac{\sqrt{3}}{2}$ **a**

Inradius = r = $\frac{a}{2\sqrt{3}}$

Circumradius = R = $\frac{a}{\sqrt{3}}$

Isosceles triangle

Equal sides are ‘a’ and other side ‘b’

Perimetre = 2a + b

Height = $\frac{1}{2}\sqrt{4a^2 - b^2}$

Area = $\frac{b}{4}\sqrt{4a^2 - b^2}$

Right angle triangle

Area $= \frac{1}{2} \times$ **base** $\times$ **height**

Relation between sides $= \quad side^2 + side^2 = hypotenuse^2$

Scalene triangle

Sides are ‘a’, ‘b’ and ‘c’

Perimetre = a + b + c

Area = $\sqrt{s(s-a)(s-b)(s-c)}$

S = semi perimetre = $\frac{a+b+c}{2}$

Rectangle

Length = l   Breadth = b

Perimetre of Rectangle = 2 (l + b)

Diagonal = $\sqrt{l^2 + b^2}$

Area = length × Breadth = lb

Square

Side = a

Perimetre = 4a

Diagonal = $\sqrt{2}$ **a**

Area = a^2

Parallelogram

Sides are ‘a’ and ‘b’

Perimetre = 2 (a + b)

Area = base × height

Rhombus

Side = a

Perimetre = 4a

d_1 and d_2 are diagonals

Area $= \frac{1}{2} \times d_1 \times d_2$

Side $= \frac{1}{2}\sqrt{{d_1}^2 + {d_2}^2}$

Trapezium

Area = $\frac{1}{2}$ (sum of parallel sides) × distance between them

Circle

Radius = r

Circumference = 2πr

Area = π r^2

Π = $\frac{22}{7}$ or 3.14

Area, if circumference is known = $\frac{c^2}{4\pi}$

Semi circle

Circumference = πr

Perimetre = πr + 2r = $\frac{36}{7}$ **r**

Area = $\frac{1}{2}\pi r^2$